DoorSensor

This door sensor will work with CoAP and HTTP. A CoAP server runs in the background and stores some resource information for our door. Then a Flask application will get the info on the door and send it in JSON format to the frontend for displaying.

This project was done to show how an HTTP request can be used in place of CoAP to serve web browsers useful content related to an IoT device.

Environment Settings

This project needs a Python compiler on your machine and a USB port for the Arduino UNO. The Arduino will connect to the computer via a USB port. The USB port's serial COM value can be found using device manager->ports on Windows OS. All 3 parts of the project, CoAPServer.py, ArduinoListener.py, and main.py, all need their own terminals open to run a specific process. These processes will run in the background and be terminated with a ctrl^C command. To see results, access localhost:8080 on your web browser after starting all three in the order I gave them to you or by following the instructions.

How to run the code

Instructions to set up and see changes from “Open” to “Closed”:

- Open a terminal and start CoAPServer.py and let it run.
- Open a terminal and start ArduinoLister.py and make sure the proper Arduino UNO is plugged into a USB or serial port like COM4 on Windows or /dev/usb on MacOS or Linux. The ArduinoListener will print out a message saying that it has received content from the Arduino UNO. The Arduino listener will make a PUT request whenever it receives relevant information.
- Open a terminal and start main.py. Some text will print out on this terminal telling you the process is running.
- Go to localhost:8080. Once you open the page there is just some generic text that says "Your door is {{/door info}}". We get the info with a GET operation. The door's status will either show "Open" or "Closed". If the door is “Open” there is a button that can be clicked, but it is not yet implemented.
- Separate or join the reed switch and magnet then refresh the page to see a change in the resource.

Instructions to activate the LED and alert the door after starting up:

- Put the magnet and reed switch next to each other to enter the “Closed” state.
- Click the button that save arm door and wait a second for the page to refresh
- Remove the magnet from the reed switch.
- This will activate the door and turn an LED on until the door is closed again.

How to interpret the results

You will be able to see the results by going to the localhost:8080 page with your web browser. The page will display the current status of your sensor and whether it is open or closed. An open signal will display some more information to tell the user the sensor's reed switch has been opened as well as the door. The website will keep you up to date given that you refresh the page every so often.

To see changes you must separate or join the magnet and reed switch then refresh the page.

Other important information

Make sure the serial port value is correct in the ArduinoListener.py. If the serial port is incorrect you will not be able to read changes in the sensor’s state. The name of the serial port also changes depending on the OS of your computer. For a windows machine you can use device manager to look at this value. I do not know how it works for a Unix/Linux system on MacOS or a Linux machine.